

OSCAR DAVIS

oscar.davis@cs.ox.ac.uk $\diamond$ [olsdavis.github.io](https://github.com/olsdavis)

Research statement. I develop generative models based on flow matching and diffusion, with a focus on continuous diffusion for language modelling and few-step sampling via flow maps.

EDUCATION

- PhD in Computer Science**, University of Oxford Oct 2023 – (Expected) Dec 2026
• Supervised by M. Bronstein, İ. Ceylan, J. Bose.
- MSc in Advanced Computer Science**, University of Oxford Oct 2022 – Aug 2023
• Supervised by M. Bronstein and İ. Ceylan. Distinction. Best dissertation prize (see below).
- Visiting Student**, Imperial College London Sep 2021 – Jul 2022
• Supervised by A. Gervais. Funded by Swiss scholarship. Distinction.
- BSc in Computer Science**, EPFL Sep 2019 – Jul 2022

RESEARCH EXPERIENCE

- Research Intern at Apple MLR**, with M. Cuturi Mar 2026 – Present
• Scaling Categorical Flow Maps to 1.7B parameters, 2.1T tokens. First continuous flow model of that scale for language modelling. Scaled self-distillation objectives, enabling coherent sampling with as few as 4 NFEs.
• Designing and analysing new variants of continuous diffusion models for language modelling.
- Research Intern at Genesis Molecular AI**, with J. Bose, N. Boffi, M. Al-Shedivat Jan – Mar 2026
• Flow maps, steering and tilting for protein generation.
- Research Intern at Microsoft Research, Cambridge**, with J. Gladrow & K. Kalinin Nov 2023 – Feb 2024
• Engineering diffusion models, latent diffusion models, VAEs, simple video models, neural ODEs.
• Theoretical analyses of Diffusion Models via SDEs, PDEs and fixed-points. (*Patented.*)

RESEARCH PROJECTS

- MSc Dissertation**, Information Theory for GNNs, with İ. Ceylan, M. Bronstein Feb – Aug 2023
• Developed a formal information-theoretic framework to fully characterise informational bottlenecks in Graph Neural Networks, including over-smoothing and over-squashing. **Received the Tony Hoare Prize for the best dissertation of the year.**
- BSc Research Project**, DeFi analysis, with A. Gervais Jan – Aug 2022
• Analysed DeFi markets on Ethereum and BNB Chain, quantifying offered financial security; implemented a custom CUDA version of the Bellman-Ford algorithm to detect arbitrage opportunities $864\times$ faster than previous SOTA.
- Student Research Project**, Scala 3.0 Compiler Extension, with M. Odersky Jun – Sep 2021
• Contributed to the thread-safe re-implementation of `lazy vals`, in the Scala 3 compiler.

PUBLICATIONS

- Efficient Reasoning with Flow Language Models** Jul 2026
Bai, H., Izermine, F., Mishra, S., [Davis, O.](#), Rusch, T.K.
COLM 2026 Efficient Reasoning Workshop.
- Scaling Categorical Flow Maps** May 2026
[Davis, O.](#), Filippova, A., Ablin, P., Turrisi, V., Shidani, A., Cuturi, M., Béthune, L.
Pre-print. [arXiv: arxiv.org/abs/2605.07820](https://arxiv.org/abs/2605.07820).

Categorical Flow Maps	Jan 2026
Roos, D.*, Davis, O.* , Eijkelboom, F.*, Bronstein, M., Welling, M., Ceylan, I., Ambrogioni, L., vd Meent, J.W. ICML 2026. arXiv: arxiv.org/abs/2602.12233 . GitHub: github.com/olsdavis/semicat .	
Generalised Flow Maps for Few-Step Generative Modelling on Riemannian Manifolds	Sep 2025
Davis, O. , Boffi, N., Albergo, M., Bronstein, M., Bose, J. ICLR 2026. NeurIPS 2025 FPI. arXiv: arxiv.org/abs/2510.21608 . GitHub: github.com/olsdavis/gfm .	
SOAPIA: Siamese-Guided Generation of Off Target-Avoiding Protein Interactions [...]	May 2025
Vincoff, S.*, Davis, O.* , Tong, A., Bose, J., Chatterjee, P. ICML 2025 FM4LS. OpenReview: openreview.net/pdf?id=Ax25SLIDsN .	
FORT: Forward-Only Regression Training of Normalizing Flows	May 2025
Rehman, D., Davis, O. , Lu, J., Tang, J., Bronstein, M., Bengio, Y., Tong, A., Bose, J. ICLR 2026. ICML 2025 GenBio (Best paper award). arXiv: arxiv.org/abs/2506.01158	
SOAPI: Siamese-guided Generation of Off-Target-Avoiding Protein Interactions	Mar 2025
Vincoff, S., Davis, O. , Tong, A., Bose, J., Chatterjee, P. ICLR 2025 GEM (Spotlight). OpenReview: openreview.net/pdf?id=aRrXs2cVdy .	
Fisher Flow Matching for Generative Modelling over Discrete Data	May 2024
Davis, O. , Kessler, S., Petrache, M., Ceylan, I., Bronstein, M., Bose, J. NeurIPS 2024. arXiv: arxiv.org/abs/2405.14664 . GitHub: github.com/olsdavis/fisher-flow .	

TEACHING EXPERIENCE

Co-Lead Teaching Assistant for generative modelling at EEML 2025 , Sarajevo	Jul 2025
• Writing and presenting a geometric generative modelling tutorial (flow matching, Riemannian flow matching).	
Graduate Teaching and Research Scholarship in CS , Oriel College, Oxford	Apr 2025 – Mar 2026
• Teaching undergraduate-level courses to students of Oriel College. Admissions interviews.	
TA for Geometric Deep Learning , University of Oxford, under M. Bronstein	Jan – Mar 2025 and 2026
• Teaching PyTorch implementations of geometric models (equi-/invariance) and others (e.g., neural diffusion).	
OÄW Winter AI School 2025 , Vienna	Jan 2025
• Gave two PyTorch tutorials: one on implementing Graph Neural Networks; one on (Riemannian) flow matching.	
TA for Graph Representation Learning , University of Oxford, under İ Ceylan	Oct – Dec 2023 and 2024
• Teaching PyTorch and PyTorch Geometric (for Graph Neural Networks, and Knowledge Graph Learning).	
TA for Object-Oriented Programming (Java) , EPFL, under M. Schinz	Feb – Jun 2020

AWARDS & SERVICE

ICML 2025 GenBio – Best paper award	May 2025
NeurIPS Top Reviewer	2025
G-Research Grant for PhD Students and Postdocs (£1k)	Feb 2024
Tony Hoare Prize for the best MSc Dissertation , University of Oxford	Sep 2023
Swiss Study Foundation Scholarship (for near 100% GPA in my final term)	Sep 2021

SKILLS

Programming	Python, PyTorch, JAX (Equinox, Grain, Orbax), Java, Go, Scala
Systems	Git, Linux, multi-GPU and multi-node distributed training